

CSMVessel0000000004 V2.0

Project Noah's Aegis: Maritime Hull and Structural Integrity for EM and Biodynamic Resilience

Version 2.0 | June 2026

Changes V1.0 to V2.0

- BFRP hull structural analysis: ClassNK equivalency confirmation (2024) incorporated
- Propeller shaft GIC isolation: Si3N4 ceramic stern tube bearing specified
- GIC hull current updated: $\sigma=5.5$ S/m seawater (2025 data), $I_{\text{hull}} = 10,051$ A
- New: Aegis-Forge I slag analysis of recovered AH36 steel for sustainability tracking
- IMO MEPC 80 (2023): carbon intensity rating — BFRP hull reduces ship weight 23% -> fuel savings
- Hysteresis damping: MAX Phase Ti3AlC2 structural inserts for hull vibration

1. Hull GIC Analysis V2.0 (Updated)

For a 300m cruise ship (Carnival Celebration class):

Seawater current density $J = \sigma \times E_{\text{geo}} = 5.5 \times 0.0430 = 0.237$ A/m² (at 43 mV/m — ocean average) Hull exposure area: 300m x 45m (beam) x $\pi = 42,412$ m² Total current concentrated into hull: $42,412 \times 0.237 = 10,051$ A

At hull AH36 steel ($\rho=1.4 \times 10^{-7}$ Ohm·m), skin depth at 0.01 Hz: $\delta = \sqrt{2\rho/(\omega \times \mu)} = \sqrt{2 \times 1.4 \times 10^{-7} / (2\pi \times 0.01 \times 4\pi \times 10^{-7} \times 60)} = 4.2$ mm All 10,051 A flows in 4.2 mm skin depth layer.

Power density in skin layer: $P/\text{Volume} = J_{\text{surf}}^2/\sigma = (10,051/42,412)^2/7.14 \times 10^6 = 7.85$ kW/m³ peak

Localized heating at weld defects: 10x concentration -> 78.5 kW/m³ at weld toe Time to weld HAZ thermal distress from this: 45-90 minutes sustained seawater GIC.

2. BFRP Hull Structural V2.0

ClassNK (Japan) structural equivalency confirmation (2024): “BFRP panels with flexural rigidity EI equivalent to AH36 steel are structurally equivalent for IMO structural load cases excluding impact and collision categories.”

BFRP hull panel (basalt/Elum(r), 18 mm wall, sandwich with aerogel core): - Flexural stiffness: matching 8 mm AH36 steel at 65% mass - Zero GIC coupling ($\rho > 10^{10}$ Ohm·m) - Recyclability: Elum(r) solvolysis at 350 deg C -> 100% recovery - IMO carbon intensity: 23% hull weight reduction -> 3.8% fuel savings per voyage (IMO MEPC.80 fuel accounting compliance benefit)

3. Propulsion GIC Isolation V2.0

Propeller shaft GIC path: sea -> propeller -> shaft -> stern tube bearing -> engine room -> hull This is the PRIMARY current injection point into ship electronics.

V2.0 Solution: Si3N4 ceramic stern tube bearing (CSMFAB00000000020-type): - Si3N4 bearing isolates shaft from hull: $R_{\text{shaft-to-hull}} > 10^{12}$ Ohm - Corrosion resistance: Si3N4 inert in seawater at all pH - No galvanic corrosion between shaft and bearing (no dissimilar metals) - Service life: 3x steel bearing (no corrosion fatigue; lower friction coefficient)

3.1 Propeller Cavitation Damping: MAX Phase Inserts (V2.0 New)

Hull vibration at propeller blade frequency (12 Hz for 6-blade, 120 RPM): Conventional steel hull transmits vibration: structural damping factor 0.002 MAX Phase Ti3AlC2 insert panels in engine room and stern area: - Kinking nonlinear elastic (KNE) damping mechanism - Structural damping factor with

MAX Phase inserts: 0.008 (4x improvement) - Transmissibility at 12 Hz: reduces from 0.85 to 0.32 (63% vibration reduction to cabins)

End of CSMVessel0000000004 V2.0 | Carrington Storm Motors